

Savitribai Phule Pune University
T.Y.BCA (SEM-V) NEP
Practical Examination OCT / NOV
CA-305-MJP: Fundamentals of Data Science

Duration: 3 Hours

Maximum Marks: 35

Q1) A Create a DataFrame from a dictionary containing employee name, [10 marks]
department, designation, and salary for at least 8 employees. Display only
the first 5 records using a suitable Pandas function.

Q2) B Write a program to create Dataset Name: StudentData.csv [20 marks]
Save the following data in Excel and store it as a .CSV file.

Name	Address	Age	Class
Smaran	Pune	20	BE
Sachin	Mumbai	23	BSc(CS)
Samarpan	Delhi	19	BVoC
Spandan	Chennai	21	BCom
Shravan	Madurai	22	BSc

Import Dataset

The dataset contains two categorical columns:

- Name
- Class

Perform:

- a) One-Hot Encoding on **Name** column
- b) Label Encoding on **Class** column

OR

Q2) B Write a program to generate a random array of 25 integers and display [20 marks]
them using a line chart, scatter plot, histogram and box plot. Apply
appropriate color, labels and styling options

Q3) Viva [05 marks]

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- Q1) A** Write a Python program using Pandas to calculate the sum, average, [10 marks]
median, mode, minimum, maximum, and standard deviation of the
following marks:

Marks = [72, 85, 91, 68, 85, 79, 94]

- Q2) B** Write a program to create two lists, one representing student names and [20 marks]
the other representing percentage. Display the data in a pie chart and bar
chart.

OR

- Q2) B** Write a program to create a DataFrame containing 2D coordinates of [20 marks]
multiple data points. Write a Python program to select any two points and
compute the Euclidean distance between them using
numpy.linalg.norm(), showing both the difference vector and the final
distance.

- Q3)** Viva [05 marks]

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Q1) A Create a DataFrame containing student names, attendance percentages, and scores, with some missing values and at least one duplicate row. Use `fillna()` to replace missing scores with the mean score and use `duplicate()` to identify duplicate records. [10 marks]

Q2) B Write a program to generate a random array of 25 integers and display them using a line chart, scatter plot, histogram and box plot. Apply appropriate color, labels and styling options [20 marks]

OR

Q2) B Write a program to create a NumPy array of the ages of 15 employees and write a Python program to compute and display the histogram using custom bins, showing the frequency of ages in each bin and plot the histogram using matplotlib with labelled axes. [20 marks]

Q3) Viva [05 marks]

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- Q1) A** Create a DataFrame from a dictionary containing product name, brand, category and price for at least 8 products. Display only the first 5 records using a suitable Pandas function. [10 marks]
- Q2) B** Write a program to create a DataFrame containing the salaries (in thousands) of 15 employees in a company. Write a Python program to calculate the quartiles (Q1, Q2, Q3) and selected percentiles (25th, 50th, 75th, 90th) and detect outliers using both the IQR and Z-score methods. [20 marks]

OR

- Q2) B** Write a program to create Dataset Name: StudentData.csv [20 marks]
Save the following data in Excel and store it as a .CSV file.

Name	Address	Age	Class
Shrikrishna	Trivendram	20	BE
Shridhar	Madurai		BSc(CA)
Mohan	Suchindram	19	BVoC
Govind	Rameswaram	21	BCom
Vasudev	Kanyakumari	22	BSc

Import Dataset

Replace missing values of:

- Age column → with mean age

- Q3)** Viva [05 marks]

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Q1) A Create a dictionary containing a list of laptop models and their corresponding prices. Use the keys Laptop_Model and Cost, convert the dictionary into a Pandas DataFrame and display it. [10 marks]

Q2) B Write a program to create two lists, one representing student names and the other representing percentage. Display the data in a pie chart and bar chart. [20 marks]

OR

Q2) B Write a program to create a DataFrame containing the monthly sales (in Lakh) of a Supply Chain Management System for all months. Write a Python program to calculate range, variance and standard deviation of sales data. [20 marks]

Q3) Viva [05 marks]

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- Q1) A** Create a DataFrame from a dictionary containing student name, age, course and marks for at least 8 students. Display only the first 5 records using a suitable Pandas function. [10 marks]
- Q2) B**
1. To create a dataframe containing columns emp_name, age and salary. Add 10 rows to the dataframe. [20 marks]
 2. Apply shape (), describe () on given frame
 3. To Add 5 rows with duplicate values and missing values. Add a column 'Department' with empty values. Display the data.

OR

- Q2) B** Write a program to create Dataset Name: StudentData.csv [20 marks]
- Save the following data in Excel and store it as a .CSV file.

Name	Address	Age	Class
Smaran	Pune	20	ME
Sachin	Mumbai	23	MSc (CA)
Samarpan	Delhi	19	MVoC
Spandan	Chennai	21	MCom
Shaman	Madurai	22	MSc

Import Dataset and Perform Basic Exploration

- a) Display statistical description of dataset
- b) Display shape of dataset
- c) Display first two rows

- Q3)** Viva [05 marks]

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- Q1) A** Create a file named staff.csv containing staff_id, employee_name, department, and monthly_salary for at least 5 employees. Read the CSV file using Pandas and display it as a DataFrame. [10 marks]
- Q2) B** Write a program to create a NumPy array of the ages of 15 employees and write a Python program to compute and display the histogram using custom bins, showing the frequency of ages in each bin and plot the histogram using matplotlib with labelled axes. [20 marks]

OR

- Q2) B** Write a program to create a student DataFrame which contains Roll No, Name and Percentage. Write a Python program to calculate mean, median and mode of given data. [20 marks]
- Q3)** Viva [05 marks]

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Q1) A Create a DataFrame from a dictionary containing book title, author, category and rating for at least 8 books. Display only the first 5 records using a suitable Pandas function. [10 marks]

Q2) B Write a program to create Dataset Name: StudentData.csv [20 marks]
Save the following data in Excel and store it as a .CSV file.

Name	Address	Age	Class
Shyama	Pune	20	BSc (Chemistry)
Shreeji	Mumbai	19	BSc (Physics)
Ramya	Delhi		BSc (Biology)
Radha	Mumbai	21	BSc (Zoology)
Satya	Pune	22	BSc (CS)

Import Dataset

Replace missing values of:

- Age column → with mean age
- Address → with One hot encoding

OR

Q2) B Write a program to implement create a DataFrame containing the salaries (in thousands) of 15 employees in a company. Write a Python program to calculate the quartiles (Q1, Q2, Q3) and selected percentiles (25th, 50th, 75th, 90th), and detect outliers using both the IQR and Z-score methods. [20 marks]

Q3) Viva [05 marks]

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Maximum Marks: 35

- Q1) A** Write a Python program to find the frequency of the three species of the Iris data (Use iris.csv) [10 marks]
- Q2) B** Generate a random array of 50 integers and display them using a line chart, scatter plot, histogram and box plot. Apply appropriate color, labels and styling options. [20 marks]

OR

- Q2) B** Write a Python program to compute complete descriptive statistics for the numeric columns (Salary) using pandas, including mean, median, variance, standard deviation, minimum, maximum, quartiles, and count. [Use employee.csv] [20 marks]
- Q3)** Viva [05 marks]

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Q1) A Create a student DataFrame which contains student_id, name and marks. Write a Python program to calculate mean, median, and mode of given data. [10 marks]

Q2) B Write a Python program [20 marks]

1. To create a dataframe containing columns emp_name, age and salary. Add 10 rows to the dataframe.
2. Apply shape (), describe () on given frame
3. To Add 5 rows with duplicate values and missing values. Add a column 'Department' with empty values. Display the data.

OR

Q2) B Write a python program to apply discretization on the continuous Marks attribute into categorical performance levels such as *Poor, Average, Good, and Excellent*. Display the dataset before and after discretization. [20 marks]
[Use Student_Marks.csv]

Q3) Viva [05 marks]

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Q1) A Prepare a DataFrame containing the monthly sales (in thousands) of a company for 12 months. Write a Python program to calculate range, variance, and standard deviation of sales data. [10 marks]

Q2) B Import dataset “iris.csv”. Write a Python program to create a Bar plot to get the frequency of the three species of the Iris data. Prepare a histogram of the three species of the Iris data. [20 marks]

OR

Q2) B Write a Python program to generate a box plot to show the Interquartile range and outliers for the three species for each feature [Use Iris.csv]. [20 marks]

Q3) Viva [05 marks]

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- Q1) A** Write a Python program to create box plots to see how each feature i.e. Sepal Length, Sepal Width, Petal Length, Petal Width are distributed across the three species. [Use iris.csv dataset] [10 marks]
- Q2) B** Write a Python program to handle missing values on Age and Fees using their mean. Encode 'City' using One-Hot Encoding and 'Admitted' using Label Encoding. [Use StudentData.csv with attributes Student name, City, Age, Fees, Admitted] [20 marks]

OR

- Q2) B** Generate a random array of 50 integers and display them using a line chart, scatter plot, histogram and box plot. Apply appropriate color, labels and styling options. Add two outliers to the above data and display the box plot. [20 marks]
- Q3)** Viva [05 marks]

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- Q1) A** Write a Python program to perform basic data exploration by displaying the shape, size, data types, statistical details, number of observations, and missing values in a dataset. [10 marks]
- Q2) B** Create a dataset data.csv having two categorical columns (the country column, and the purchased column). [15] [20 marks]
- a. Apply OneHot coding on Country column.
 - b. Apply Label encoding on purchased column

OR

- Q2) B** Write a Python program to generate 50 random integers and visualize them using a line chart, scatter plot, histogram and box plot with suitable labels and styling. Then add two outliers and display the updated box plot. [20 marks]
- Q3)** Viva [05 marks]

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Q1) A Create a dictionary that stores student names as values for the key '**Name**', marks as values for the key '**Marks**', and city names as values for the key '**City**'. Convert the dictionary into a DataFrame and display it. [10 marks]

Q2) B Write a Python program to compute complete descriptive statistics for the numeric columns (Salary) using pandas including mean, median, variance, standard deviation, minimum, maximum, quartiles and count. [20 marks]
[Use employee.csv]

OR

Q2) B Import dataset "iris.csv". Write a Python program to create a Bar plot to get the frequency of the three species of the Iris data. Write a Python program to create a histogram of the three species of the Iris data. [20 marks]

Q3) Viva [05 marks]

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- Q1) A** Write a python program to create two lists, one representing subject names and the other representing marks obtained in those subjects. Display the data in a pie chart and bar chart. [10 marks]
- Q2) B** Write a Python program to read a CSV file named data.csv and perform following operations [20 marks]
- Display DataFrame
 - describe ()
 - head ()
 - tail ()
 - max ()
 - min ()

OR

- Q2) B** Write a Python program for Handling Missing Value. Replace missing value of salary, age column with mean of that column. Generate a line plot of name Vs salary [Use Data.csv file]. [20 marks]
- Q3)** Viva [05 marks]

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- Q1) A** Create a student (rollno, name, mark1, mark2, mark3), dataframe. Apply max(), min(), describe(), shape() [10 marks]
- Q2) B** Write a Python program that takes a dataset of experiments and computes **weighted arithmetic, harmonic and geometric means** along any chosen axis. Use vectorized NumPy operations and allows custom weights. [20 marks]

OR

- Q2) B** Create a dataset data.csv having two categorical columns (the country column, and the purchased column). [20 marks]
a. Apply OneHot coding on Country column.
b. Apply Label encoding on purchased column
- Q3)** Viva [05 marks]

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Q1) A Create a NumPy array containing the ages of 10 employees. Write a Python program to calculate and display the minimum age, maximum age and average age. [10 marks]

Q2) B Download any dataset from the UCI repository. Read the CSV file using read_csv() function. Describe the dataset using an appropriate function. Display the mean value of numeric attributes and check whether any data values are missing. [20 marks]

OR

Q2) B Create an Employee DataFrame containing Employee_ID, Department, Age and Salary. Implement Simple Random Sampling by selecting 5 employees, Systematic Sampling by selecting every 3rd employee, and Stratified Sampling by grouping employees according to Department and selecting one employee from each department. [20 marks]

Q3) Viva [05 marks]

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Q1) A Create a NumPy array of 10 numerical values and write a Python program to calculate and display its sum, mean, minimum and maximum. [10 marks]

Q2) B Download the Iris dataset and read it using `read_csv()`. Take samples from the complete dataset. Display the maximum and minimum values of all numerical attributes and find the number of records for each distinct value of the class attribute. [20 marks]

OR

Q2) B Create a DataFrame containing 2D coordinates of multiple data points. Select any two points and calculate both Euclidean distance and Manhattan distance. Display the difference vector and the calculated distances. [20 marks]

Q3) Viva [05 marks]

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Maximum Marks: 35

Q1) A Create a NumPy array containing the marks of 10 students and write a Python program to sort the marks and display the highest three marks. [10 marks]

Q2) B Write a Python program that takes a dataset of experiments and computes weighted arithmetic mean, weighted harmonic mean and weighted geometric mean along a selected axis. Use custom weights. [20 marks]

OR

Q2) B Write a Python program to compute the normalized cross-correlation between two multidimensional arrays along a specified axis and report the lag of maximum correlation. [20 marks]

Q3) Viva [05 marks]

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Q1) A Create a NumPy array containing numerical data and write a Python program to calculate its variance and standard deviation. [10 marks]

Q2) B Write a Python program to identify outliers in a dataset using IQR, Z-score and Modified Z-score methods. Analyse how removing the detected outliers affects the mean, median and standard deviation. [20 marks]

OR

Q2) B Create a DataFrame containing salaries of 20 employees. Calculate Q1, Q2, Q3, 25th, 50th, 75th and 90th percentiles and identify outliers using the IQR method. [20 marks]

Q3) Viva [05 marks]

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Q1) A Create a NumPy array containing the marks of students and divide the marks into three groups using conditional statements. Display the number of students in each group. [10 marks]

Q2) B Generate a random array of 50 integers and display it using a line chart, scatter plot, histogram and box plot. Apply appropriate labels and styling options. Add two outliers to the data and display the updated box plot. [20 marks]

OR

Q2) B Generate two random datasets from normal distributions. Display both datasets using histograms in a single figure with suitable bins, transparency and edge styling. [20 marks]

Q3) Viva [05 marks]

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- Q1) A** Create two lists representing subject names and marks obtained. Convert the lists into a DataFrame and display the DataFrame. [10 marks]
- Q2) B** Write a Python program to read a CSV file named employee.csv and perform following operations [20 marks]
- Display DataFrame
 - describe ()
 - head ()
 - tail ()
 - max ()
 - min()

OR

- Q2) B** Create a Matplotlib Figure and Axes object for a given numerical dataset. Set the figure size and customize the title, x-label, y-label, line width and other suitable properties. [20 marks]
- Q3)** Viva [05 marks]

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Q1) A Create a DataFrame containing five products and their prices. Display the DataFrame and calculate the maximum, minimum and average price. [10 marks]

Q2) B Create a numerical dataset and apply L1 normalization so that the sum of absolute values in each row becomes 1. Display the original and normalized data. [20 marks]

OR

Q2) B Import diabetes.csv and perform Min-Max rescaling, standardization and unit-length normalization on its numerical attributes. Display the transformed data. [20 marks]

Q3) Viva [05 marks]

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Q1) A Create a DataFrame containing student names, study hours and marks. Display the first three rows, last three rows and the shape of the DataFrame. [10 marks]

Q2) B Import diabetes.csv and apply binarization to the numerical attributes using a suitable threshold. Display the original and binary data. [20 marks]

OR

Q2) B Import Student_Marks.csv and discretize the continuous Marks attribute into the following performance levels: Poor, Average, Good and Excellent. Display the dataset before and after discretization. [20 marks]

Q3) Viva [05 marks]

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- Q1) A** Create a DataFrame containing product names, quantities and prices. Display the DataFrame and calculate the total quantity and average price. [10 marks]
- Q2) B** Create a DataFrame containing product prices and perform equal-width binning to divide the prices into three categories: Low, Medium and High. Display the original and binned data. [20 marks]

OR

- Q2) B** Create a numerical dataset and perform unit-length normalization on the data. Display the original data and the normalized values. [20 marks]
- Q3)** Viva [05 marks]

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Q1) A Create a NumPy array containing numerical values and standardize it using `preprocessing.scale()`. Display the original and standardized values. [10 marks]

Q2) B Using `iris.csv`, create a box plot to show how Sepal Length, Sepal Width, Petal Length and Petal Width are distributed across the three species. [20 marks]

OR

Q2) B Using `iris.csv`, create a Seaborn pair plot and analyse the relationships among the different features. Identify which flower species appears to be the most separable. [20 marks]

Q3) Viva [05 marks]

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Q1) A Create a DataFrame containing five categories and their corresponding values. Display the DataFrame and calculate the maximum and minimum values. [10 marks]

Q2) B Using iris.csv, create a Seaborn joint plot with KDE to show the bivariate distribution between Sepal Length and Sepal Width. [20 marks]

OR

Q2) B Using iris.csv, create box plots to show the Interquartile Range and outliers for the three species for each feature. [20 marks]

Q3) Viva [05 marks]